

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in production RAG systems, custom AI agents, and enterprise-grade LLM workflows for regulated healthcare environments. Proven track record deploying Claude-integrated multimodal systems (Manga Lens), building multi-agent pipelines with schema-validated reasoning contracts, and achieving measurable production outcomes: ~35% contextual retrieval precision improvement, >30% hallucination reduction, and >90% retrieval-grounded response alignment. Experienced with advanced prompt engineering, agent evaluation frameworks, and HIPAA-conscious data governance - directly aligned with Anthropic's regulated-industry enterprise focus.

CORE COMPETENCIES

LLM Agent Development • RAG System Design • Prompt Engineering & Evaluation • MCP/Sub-Agent Orchestration • Healthcare AI (HIPAA-conscious) • Python & TypeScript • LangChain & LlamaIndex • FastAPI / Docker • Hallucination Guardrails • Enterprise AI Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **production RAG systems** for clinical knowledge retrieval using recursive semantic chunking and transformer embeddings; improved contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, and achieved >**90%** retrieval-grounded response alignment - directly applicable to enterprise AI agent deployment in regulated industries.
- Developed **custom agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%** and reduced hallucinations >**30%**.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker, with structured logging and load simulation; reduced post-deployment defects ~**30%**.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs - aligned with Anthropic's regulated-industry enterprise requirements.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient risk scoring; improved recall **15-20%** on high-risk categories while holding precision >**90%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP | Hyderabad, India

- Built **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; reduced manual review time **60-70%** and achieved ~**85% alignment** with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%** and raised contextual answer accuracy >**85%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Built CI/CD pipelines with Jenkins; reduced deployment errors >**30%** and improved release cycle time ~**35-40%**.
- Optimized SQL/T-SQL stored procedures; reduced query execution time ~**20%** and data retrieval latency ~**25%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, multi-agent orchestration

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Integrates **four AI vision providers including Claude Sonnet** (GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline with coordinate remapping, 7-day translation cache, per-domain selector configs for 29 sites.

Demonstrates hands-on Claude API integration in production.

TypeScript, Manifest V3, Claude Sonnet API, GPT-4o mini, multimodal AI

Agentic Pixel Character Synthesis & Animation Engine

LLM-based agent orchestrator that decomposes high-level prompts into generation tasks; Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning.

PyTorch, HuggingFace Diffusers, FastAPI, Docker, LLM agent orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, SQL

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails, Claude API, OpenAI API

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Enterprise AI, Regulated Industry Deployment, Workflow Automation